Evaluation Report of Elite_FS_Day_21_EigenVectors-KMIT

Average Score: 6/10 | Attempted: 3 of 3 questions

Question 1

THEORY UNDERSTAND Best Score: 5/10

What is Data Drift? Explain different types of data drift.

Desired Answer

Data drifting (or data drift) refers to a change in the statistical properties of input data over time compared to the data used to train a machine learning model. When the underlying data distribution shifts, the model's predictions may become less accurate because it was trained under different conditions. Formally, if a model was trained on data drawn from distribution Ptrain(X), and over time the incoming data follows a different distribution Plive(X), then: Ptrain(X)□= Plive(X) This difference is called data drift. Types of Data Drift 1. Covariate Drift (Feature Drift): The distribution of input features changes, but the relationship between input and output remains the same. 2. Concept Drift: The relationship between input features and the target variable changes over time. 3. Label Drift (Prior Probability Shift): The distribution of the target variable changes

Student's Answer

when statistical distribution of input changes over time, compared to training data. As the time passes by the data changes and the model no longer performs well. Types: Covariate Drift: Input feature distribution changes. but the relationship is same Label Drift: Distribution of the output class changes.. Mean Drift: Variance Drift:

Evaluations

Feedback 1

★★½☆☆ 5/10

The student is addressing the same core concept of data drift as the Actual Answer (a shift in the statistical distribution of incoming data compared to training data). They accurately defined data drift in their own words and correctly described two of the three types listed in the Actual Answer (Covariate Drift and Label Drift). However, they omitted Concept Drift entirely (the change in the relationship between inputs and the target) and introduced 'Mean Drift' and 'Variance Drift,' which were not part of the Actual Answer's classification. This

shows a basic but incomplete understanding of the required categories and scope, warranting an Adequate Understanding score.

Feedback 2

5/10

The student is discussing the same concept of data drift as the Actual Answer, so the response is relevant. They correctly define data drift as a change in the statistical distribution of inputs over time and note its impact on model performance. They accurately describe Covariate Drift and Label Drift. However, they omit the key concept of Concept Drift (the change in the input–output relationship), which is explicitly covered in the Actual Answer. They also list "Mean Drift" and "Variance Drift," terms not mentioned or defined in the Actual Answer, without explanation. Thus, the student shows a basic grasp of data drift and two of its types but misses a major category and adds ungrounded types, reflecting only partial understanding of the full Actual Answer.

Question 2

THEORY REMEMBER Best Score: 7/10

What is KL Divergence?

Desired Answer

Kullback–Leibler (KL) Divergence is a measure from information theory that quantifies how one probability distribution differs from a reference probability distribution. It represents the amount of information lost when an approximate distribution Q is used to represent the true distribution P. For discrete probability distributions, KL divergence is defined as: $D_{KL}(P\|Q)=\sum P(x)\log(P(x)/Q(x))$ Conceptually, KL divergence measures the expected extra number of bits required to encode samples from distribution P when using a coding scheme optimized for Q instead of P. Thus, it quantifies the divergence in information content between two probability distributions.

Student's Answer

It measures how one probability distribution differs from other $KIL(P/Q)=P(x)* \log(P(x)/Q(x))$

Evaluations

Feedback 1

★★★☆☆ 6/10

The student correctly states that KL divergence measures how one probability distribution differs from another and recalls the core form P(x)·log(P(x)/Q(x)). However, they omit the summation over x (∑ P(x) log(P(x)/Q(x))) and do not mention the information-theoretic interpretation (extra bits required or information lost). Thus, they capture some key facts (definition and formula structure) but miss important details and the conceptual context from the Actual Answer.

Feedback 2

★★★½☆ 7/10

The student correctly recalled the core idea that KL divergence measures how one probability distribution differs from another and wrote down the basic form of the formula P(x)·log(P(x)/Q(x)). However, they omitted the summation over x (∑_x) that is essential to the discrete definition, and they did not mention the conceptual interpretation of KL divergence as the extra bits or information lost when using Q to approximate P. These omissions prevent a full, excellent recall but demonstrate good recall of the primary facts.

Question 3

THEORY UNDERSTAND **Best Score: 5/10**

What is the significance of eigenvectors for an AI engineer?

Desired Answer

Eigenvectors are significant for an AI engineer because they identify the fundamental directions along which data or transformations behave most strongly. Mathematically, for a matrix A, an eigenvector v satisfies Av=λv, meaning the transformation scales the vector without changing its direction. This helps reveal the intrinsic structure of data and models. In practical AI applications, eigenvectors are central to Principal Component Analysis (PCA), where they determine the directions of maximum variance for dimensionality reduction and feature extraction. They also play a crucial role in analyzing model stability and optimization, particularly through Hessian eigenvalues, which indicate whether a model is converging to sharp or flat minima. Additionally, eigenvectors are used in spectral clustering, graph-based learning, Markov processes, and representation learning, where they uncover hidden patterns, clusters, and steady-state behaviors. Overall, eigenvectors provide a mathematical foundation for understanding data geometry, learning dynamics, and structural properties of AI systems.

Student's Answer

Eigen Vector: Av=lambda v, They help t=in knowing the inner structure of data and models, they provide mathematical foundation, in understanding the plane structure, properties of AI . Normally when a value of added to vector both direction and value changes but for eigen vector only the vector is scaled.. Along which directions does the transformations of vector typically point to

Evaluations

Feedback 1

5/10

The student is addressing the same concepts as the Actual Answer (the eigen equation Av=λv, scaling without changing direction, and that eigenvectors reveal intrinsic data structure). They correctly state the core definition and the idea that eigenvectors show important directions in data or transformations. However, their answer is very brief, lacks clarity, and does not explain the specific AI applications (PCA for dimensionality reduction, Hessian eigenanalysis for model stability, spectral clustering, etc.) or how eigenvectors underpin learning dynamics. They touch on 'inner structure' and 'mathematical foundation' but miss most of the practical contexts and deeper relationships presented in the Actual Answer. This warrants an adequate but not strong score.

Feedback 2

5/10

The student is addressing the same core concept as the Actual Answer (the eigenvector equation Av=λv, the idea that eigenvectors are scaled without direction change, and that they reveal the inner structure of data/models). However, their answer is very brief and misses most of the key applications and details from the Actual Answer: they do not mention PCA, Hessian eigenvalues for optimization, spectral clustering, Markov processes, or other AI-specific uses. They capture the basic mathematical definition and the notion of intrinsic structure, but lack depth and coverage of the primary AI engineering significance outlined in the Actual Answer.